Aswin Sanosh

AI Automation Engineer

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Professional Summary

Full Stack Developer and AI/ML Engineer with hands-on experience building web applications, real-time systems, and data-driven tools through freelance and startup work. Comfortable developing end-to-end products with Django, React.js, and modern backend frameworks, while also working on computer vision, unsupervised learning, and physics-informed AI models. Experienced in REST APIs, real-time data pipelines, and IoT integrations for projects such as stock prediction, flood forecasting, and smart transport systems. Also involved in mentoring software teams, improving system performance, and delivering practical solutions in research-focused and startup environments.

Core Skills

Programming Languages: Python, JavaScript, TypeScript, Java, C, C++, PHP, R, VB .NET, SQL, Dart
Frontend & Mobile: HTML, CSS, React.js, Next.js, Flutter, React Native, Three.js, Tailwind CSS
Backend & APIs: Django, Django REST Framework, Node.js, Express.js, NestJS
AI/ML, Data & Computer Vision: NumPy, Pandas, SciPy, Scikit-learn, TensorFlow, PyTorch, OpenCV, YOLOv8, Hugging Face Transformers, MLflow
Databases: PostgreSQL, MySQL, MongoDB, SQLite, Firebase, Supabase
Cloud, Tools & Platforms: AWS (Amplify, Lambda, EC2, S3), Azure, Git, VS Code, PyCharm, Google Colab, Figma, Blender, Arduino IDE
Domains: Full-Stack Development, AI/ML, IoT, Computer Vision

Professional Experiences

Software Mentor and Developer — Feb 2025 – Present
Tessat Space Pvt. Ltd. (Startup), Kottayam, Kerala, India
- Mentoring software team and Developing full-stack applications and conducting AI/ML research.

Intern — Dec 2025 – Mar 2026
National Chung Cheng University, Minxiong, Chiayi, Taiwan
- Digital twin design for flood inundation using physics-informed neural networks.

Web-Master — Dec 2024 – Dec 2025
IEEE Computer Society Student Chapter, Saintgits College of Engineering
- Implemented web solutions for events, announcements, and engagement.

Intern — July 2024 – Sep 2024
National Chung Cheng University, Taiwan
- Developed a stock prediction website using Django and React.js.

Significant Projects

TrueSeal
AWS EC2, Next.js, Tailwind CSS, Typescript, Django, Flask (AI backend), Celery, Redis, Redux
- Built an AI-powered recruitment platform integrating deep verification into ATS workflows to connect verified talent with data-driven businesses.

Matrimony Website
AWS Amplify, Flutter, Next.js, Tailwind CSS, Typescript, Firebase, Redux, RBAC, Oracle, Razer Pay
- Built a matrimony website for a community.

Digital Twin for Flood Inundation
Python, PyTorch, Graph Neural Networks, HEC-RAS
- Developed a physics-informed flood inundation digital twin using graph neural networks on unstructured meshes, integrating HEC-RAS hydrodynamic outputs with terrain/rainfall data and custom physical-constraint loss functions for stable, interpretable predictions.

Stock Prediction Website
React.js, Tailwind CSS Django, PostgreSQL

- Built a full-stack stock prediction platform with pattern detection, visual analytics, REST APIs, and real-time alerts.

Body Shape Classification
PhP, Python, OpenCV, OpenPose, Shapy AI

- Developed an AI-based body shape classification system using pose estimation, body geometry extraction, and anthropometric analysis for fit recommendations.

Ground Penetrating Radar (GPR) Image Classification
Python, Keras, DINOv3, DBSCAN, BIRCH, K-Means

- Combined feature extraction using DINOv3 and CNN backbones with clustering methods such as DBSCAN, BIRCH, and K-Means to improve pattern discovery and analysis quality.

Industrial Tool Classification
C++, OpenCV, HOG, Hu Moments

- Built a real-time industrial tool classification system using HOG and Hu Moments in OpenCV for manufacturing and inspection workflows.

Samyuktha 2025 Website
AWS Lambda, Next.js, Tailwind CSS, Redux, MySQL

- Built the official event platform for a college techno-cultural fest with registrations, announcements, participant dashboards, and coordination workflows.

CANSAT Flight Software
Django, Vite React.js, Tailwind CSS, MySQL, ESP32, Thermal Sensors, IoT

- Engineered flight software with telemetry dashboards, anomaly detection, and alerting workflows using ESP32-based microcontrollers and thermal sensor data for satellite operations.

Helekin Technologies Website
Next.js, Tailwind CSS, Three.js, MySQL

- Built a 3D-printing e-commerce platform with interactive model customization, order tracking, and product management workflows.

Redwills Interactive Website
Vite React.js, Tailwind CSS

- Developed an immersive studio website with interactive 3D visuals and a narrative-driven experience aligned with gaming aesthetics.

Education

Integrated Master of Computer Applications 2021 – 2026
Saintgits College of Engineering, Kottayam, Kerala, India
CGPA: 7.79/10

Certifications and Awards

- Developing Applications with SQL, Databases, and Django - Coursera
- Python for Machine Learning & Data Science - Udemy
- Electronics and PCB Design - Udemy
- Winner - Meta SparkAR Competition (April 2021)

References

Prof. Pao-Ann Hsuing, PhD,
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